

Testing the 12V Rail of an ATX Power Supply

Equipment Required

- Digital multimeter
- ATX power supply
- Paper clip or ATX PSU tester (to turn on the PSU if it is disconnected from the motherboard)

Steps

1. Turn off the computer and unplug the power supply from the wall outlet.
2. Disconnect the 24-pin ATX connector from the motherboard.
3. Start the power supply (if testing outside the PC) by connecting the green wire (PS_ON) to any black (ground) wire with a paper clip.
4. Plug the power supply back into the wall outlet and switch it on.
5. Set the multimeter to DC Voltage (20 V range or auto-range).
6. Insert the black probe into any black (ground) wire on the ATX connector.
7. Insert the red probe into a yellow (+12 V) wire.
8. Read and record the voltage displayed on the multimeter.

Item	Value
Expected 12V Rail	11.40 V – 12.60 V (ATX specification)
Measured Voltage (Example)	12.08 V DC
Status	Within normal range

Interpretation

- **11.40–12.60 V:** Normal output according to the ATX specification.
- **Below 11.5 V:** The power supply may be failing or unable to provide stable power. This can cause:

- Random computer shutdowns
 - Unexpected restarts
 - Failure to boot
 - System instability under load
- **Well below 11.4 V or fluctuating significantly:** Strong indication of a **faulty Power Supply Unit (PSU)**. It may also indicate excessive load on the PSU or a short circuit in the system, but the PSU itself is a common cause and should be tested further or replaced if confirmed faulty.